

True Tags and Predicted Tags

Sentence 10150:

those coming from other denominations will welcome the opportunity to become informed .

True Tags:

DET VERB ADP ADJ NOUN VERB VERB DET NOUN PRT VERB VERB .

Predicted Tags:

DET NOUN ADP ADJ NOUN VERB VERB DET NOUN PRT VERB VERB .

Sentence 10151:

the preparatory class is an introductory face-to-face group in which new members become acquainted with one another .

True Tags:

DET ADJ NOUN VERB DET ADJ ADJ NOUN ADP DET ADJ NOUN VERB VERB ADP NUM DET .

Predicted Tags:

DET ADJ NOUN VERB DET ADJ NOUN NOUN ADP DET ADJ NOUN VERB VERB ADP NUM NOUN .

Sentence 10151:

it provides a natural transition into the life of the local church and its organizations .

True Tags:

PRON VERB DET ADJ NOUN ADP DET NOUN ADP DET ADJ NOUN CONJ DET NOUN .

Predicted Tags:

PRON VERB DET ADJ NOUN ADP DET NOUN ADP DET ADJ NOUN CONJ DET NOUN .

For the case POS tagger does produce the correct tags:

Most of them are common words with clear part-of-speech associations. For example, the word "the" in sentence 10151 is correctly labeled as DET and the word "it" in sentence 10152 is correctly labeled as PRON. These words appear frequently in the training data and they typically have a single primary part of speech, enabling the model to learn strong associations between them and their correct parts of speech. So the model assigns high emission probabilities to these words for their correct parts of speech during training.

For the case POS tagger does not produce the correct tags:

1. "coming" in sentence 10150:

Transition Probability: The model has a significantly higher probability of transitioning from DET to NOUN than from DET to VERB. This heavily influences the model's decision towards predicting "coming" as NOUN.

Emission Probability: Although the context may slightly favor a VERB, the overwhelming transition probability towards NOUN leads to the incorrect tag.

2. "face-to-face" in sentence 10151:

Transition Probability: The probability of transitioning from ADJ to NOUN is much higher than staying on ADJ, which biases the prediction.

Emission Probability: Even though being tagged as ADJ might fit the context better based on emission probabilities, the combined effect of higher probabilities for transitioning to and being a NOUN results in the incorrect prediction.

3. "another" in sentence 10151:

Transition Probability: The transition probability from NUM to NOUN is much higher compared to DET, heavily skewing the outcome.

Emission Probability: Although the DET might be a more contextually appropriate tag based on emission probabilities, the significantly higher transition probability toward NOUN dictates the incorrect prediction.

Summary: In these instances, the Viterbi algorithm selects the tag that maximizes the product of transition and emission probabilities at each step. This approach often favors a tag with higher transition probabilities, even if the emission probabilities for the next word are slightly better under the true tag. Consequently, words that can function in multiple grammatical roles or are part of less common constructions are particularly vulnerable to misclassification if the model has learned skewed probabilities based on the training data. This explains why, in each case, the POS tagger defaults to the more statistically probable tag, even when it contradicts the contextually appropriate choice.